ASUS UGEN AI LEAGUE 2026 · BATTLEFIELD LIGHTNING · WORKPLACE AI

TRACE

From alarm noise to one verified incident — on a 2.5 W ASUS UGen300.

An edge copilot for the station duty officer: it fuses cameras, alarm panels and staff radio into evidence-linked incident cards, and wakes an on-device VLM only when a second opinion is needed.

PROPOSAL · STAGE I

WEI-LUN (ALEN) HSIAO · B13902049 · NATIONAL TAIWAN UNIVERSITY

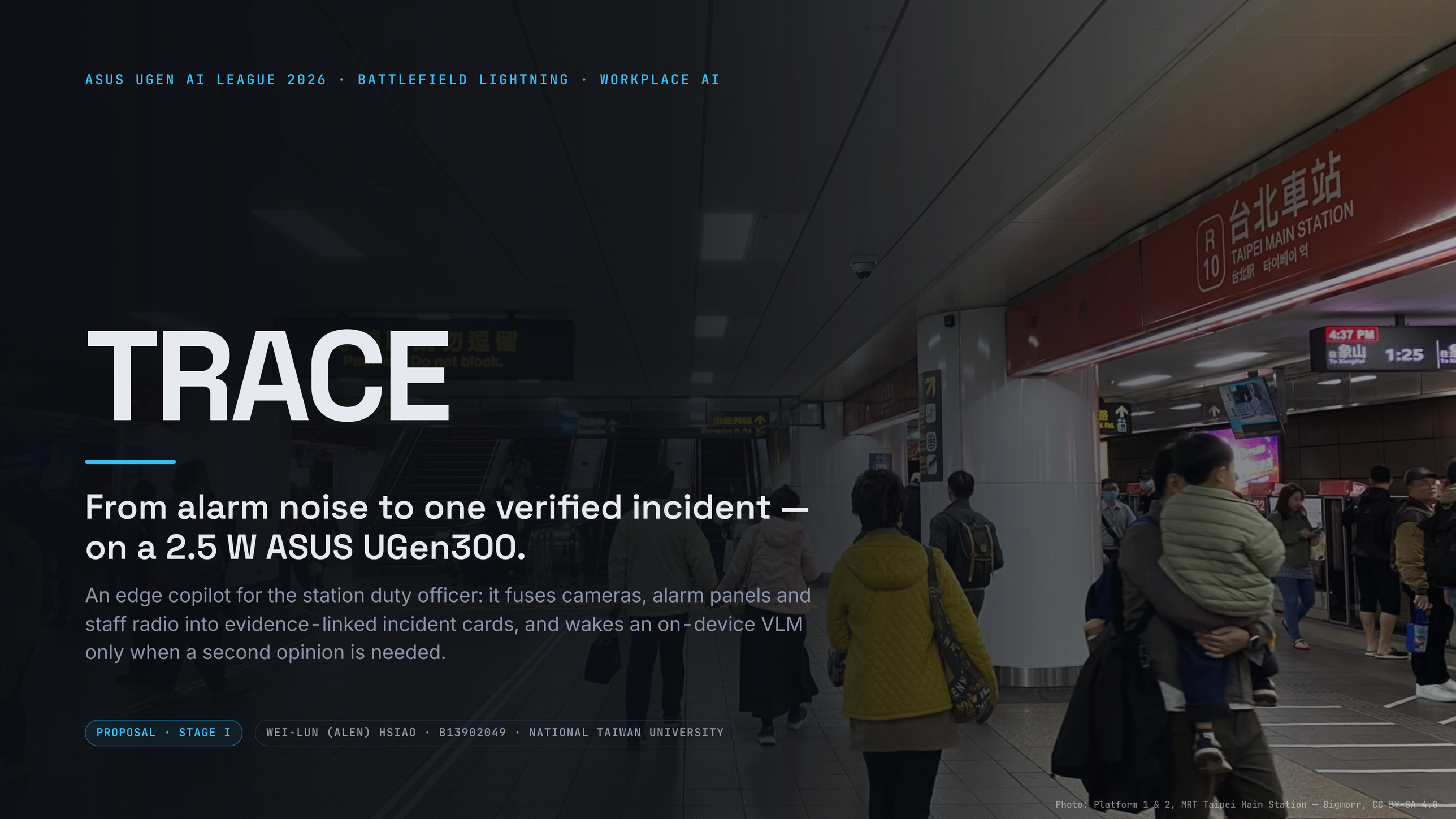


Photo: Platform 1 & 2, MRT Taipei Main Station — Bigmorr, CC BY-SA 4.0

01 / WHO WE BUILD FOR

One duty officer. Dozens of feeds. Every alarm looks the same.

At a busy station, a single officer watches camera walls, an alarm panel and a radio channel at once. Each system alerts on its own — nobody connects them.

TAIPEI METRO, 2025

2.10M

trips per day across 118 stations

AI CAMERAS BEING ADDED

280 → 2,901

Phase I → Phase II (2027–28), with new automated alert types

DEC 19, 2025

During the attack on the Taipei Metro, **41 detector groups** were activated across different zones. Operators had to reconstruct what was happening alarm by alarm. More sensors did not mean a clearer picture.



Sources: Taipei Metro ridership 2025 (Wikipedia / Taipei City Gov.); Taiwan News 2026-08-21 (AI camera rollout); Taipei Metro 1219 response report (2026-01)

Photo: People at Bannan Line platform, Taipei Main Station — Bernard McManus, CC BY 2.0

Detection scales. Human attention doesn’t.

~20 min

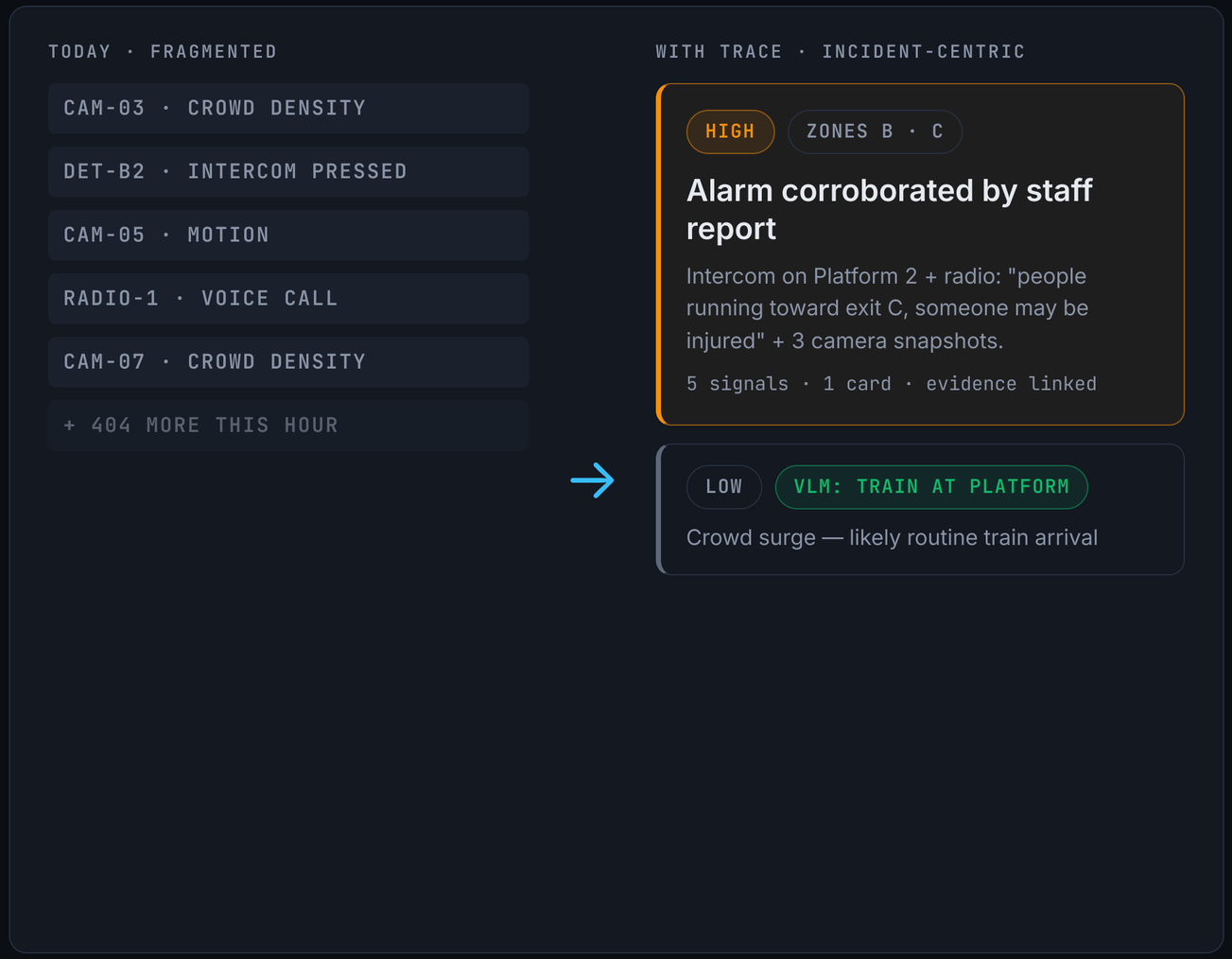
until a CCTV operator's attention falls below acceptable levels (U.S. NIJ study).

94–98%

of alarm calls are false in classic alarm systems (U.S. POP Center guide).

−9%

is all that per-sensor de-duplication — what a typical VMS offers — removes in our alert-storm test.



Sources: Green, NIJ NCJ 178265 (1999); Sampson, POP Center "False Burglar Alarms" (2007); TRACE alert-storm benchmark (synthetic, see p.15)

03 / SECOND DEPLOYMENT · THE ROAD

Same pipeline, outdoors: flag the motion, never infer the drug.

Traffic units review roadside video by hand. TRACE screens for observable risky motion and hands over a short, explainable clip.

ROAD DEATHS, 2025

2,858

~61% of H1-2025 deaths were motorcycle riders (831 of 1,368)

DRUG-DRIVING MAJOR CRASHES

90 → 266

2024 → 2025 (deaths 11 → 19)

Camera evidence can show **repeated lateral deviation** or **wrong-way travel**. It cannot show intoxication — so TRACE never labels a rider, it queues a clip for a human.



Sources: UDN 2026-03-12 (MOTC 30-day deaths); LTN 2025-09-16; National Police Agency via TVBS 2026-06-04

Photo: Taipei Bridge motorcycle waterfall — Tze Chiang Hao, CC BY-SA 4.0

04 / THE SOLUTION

TRACE: a local intelligence layer between raw signals and the officer.

It plugs into what a station already has — camera streams, the alarm panel's I/O, the radio channel — and outputs one reviewable incident per real-world event.

01 · SENSE

Always-on, cheap

YOLO + ByteTrack on every camera; alarm I/O; Whisper on staff radio. Rules emit signals: crowd surge, person down, lateral weaving…

02 · RELATE

Fuse by time + place

A zone graph groups signals that are close in time and space. Non-visual alarms pull camera snapshots on demand.

03 · VERIFY

VLM second opinion

Only for fused incidents, Qwen2-VL wakes on the same chip and asks one targeted question (e.g. "is a train at the platform?").

04 · EXPLAIN

Evidence-linked card

Masked 6-s clip, timeline, transcript, every sentence cites its signal ID, plus a recommended checklist.

05 · REVIEW

Human decides

Confirm / escalate / dismiss in one key press. Every action — including the model's — lands in an audit log.

2.5 W
accelerator power, fanless

4
model families on one chip

−98%
cards vs raw alarms, 99% recall

0
raw frames sent off the box

METRO: ALARM + RADIO + CAMERA FUSION

ROAD: TRAJECTORY RISK SCREENING

SAME EVENT SCHEMA · SAME CONSOLE · SAME BOX

One 2.5 W stick runs four model families — so the box can see, listen, verify and answer.

ASUS UGen300 · Hailo-10H
40 TOPS INT4 · 8 GB LPDDR4 · 2.5 W typ.



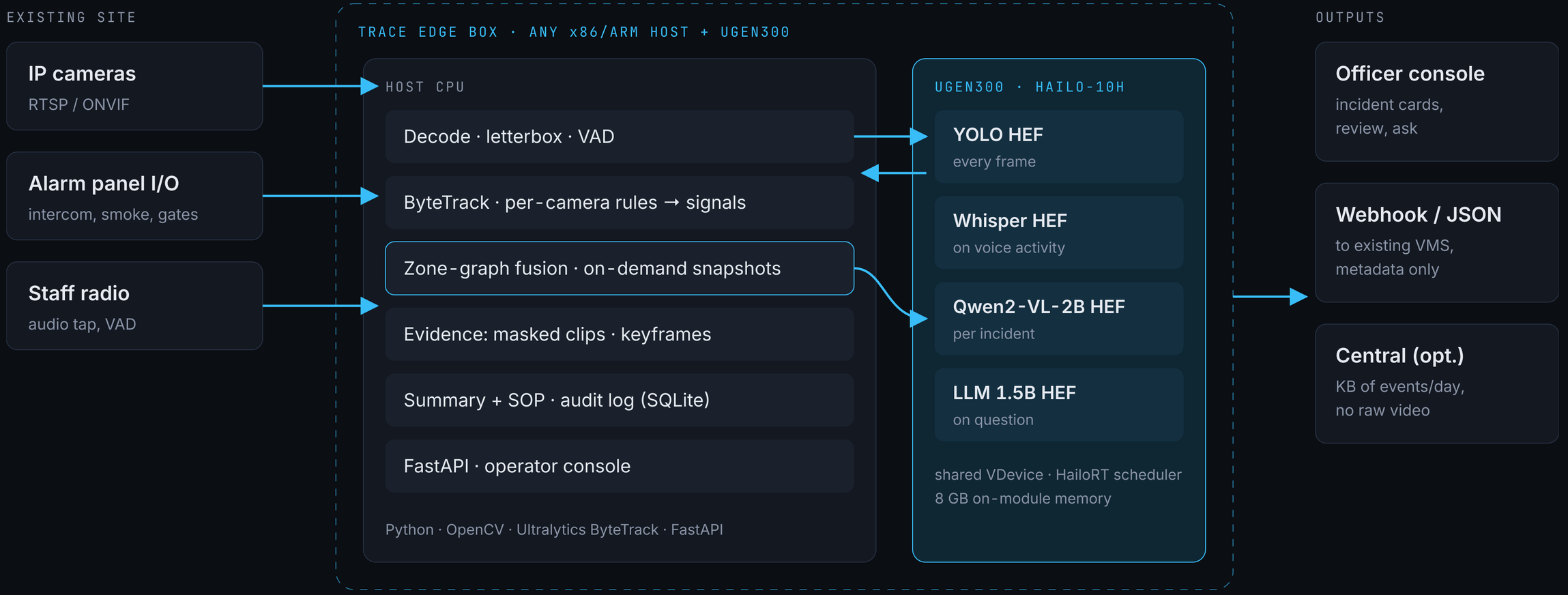
ROLE IN TRACE	MODEL (HAILO ZOO)	HAILO-10H PERF.	RUNS
Sense people & vehicles	YOLOv11n / YOLOv8s	302 / 166 FPS	always
Listen to staff radio	Whisper-Base	25 tok/s	on voice activity
Verify & describe scene	Qwen2-VL-2B	7 tok/s · TTFT 1.0 s	per incident
Answer "ask the station"	Qwen2.5-1.5B	7.4 tok/s	on demand

The cascade is the design

The cheap detector runs on every frame; the expensive models wake only on events. All HEFs share one `VDevice(group_id="TRACE")`, so HailoRT time-slices them on a single chip — no GPU, no server rack, no video leaving the station.

Sources: ASUS UGen300 press release (2026-04-01) & Sales Kit (2026-09-17); Hailo Model Zoo v5.4 (HAILO10H detection, PCIe) and GenAI Model Zoo (HailoRT 5.4). USB throughput to be measured in Stage II.

Edge-first: raw video and audio never leave the box.



Implemented today on a laptop (CPU/MPS) with identical host code; the Hailo backends are written against HailoRT 5 / hailo-apps APIs and swap in by config.

Group by time and place. Score by independent evidence.

1 · Zone graph + anchoring

Each site is a graph of zones (concourse, platform, exits). A signal joins an open incident if it is within **30 s** of its last signal and within **1 hop** of the zone where the incident started. Anchoring plus a 180 s cap stops noise from chaining a whole station into one blob.

2 · Camera-on-demand

When a non-visual alarm fires (intercom, radio), TRACE snapshots cameras in the same and adjacent zones at that moment — context for the officer, never counted as corroboration.

3 · Human report rule

A confident staff radio report plus any independent machine sensor is always **high** priority.

Score

$$\text{score} = (1 - \prod_{\text{sensors}}(1 - \hat{c}_s)) \cdot \kappa(k)$$

$$\hat{c} = \max(0, (c - 0.3) / 0.7)$$

$$\kappa(k) = \min(1, 0.70 + 0.15 \cdot (k - 1))$$

- **One vote per sensor** — a chattering camera can't inflate the score.
- **Confidence floor** — weak triggers (< 0.3) add nothing.
- **κ rewards independent source types** (camera, detector, radio): camera-only evidence is capped at medium.
- High ≥ 0.8 with ≥ 2 sensors; medium ≥ 0.55.

Code: traceai/fusion.py · unit tests: tests/test_fusion.py, tests/test_verify.py

A detector sees “more people”. The VLM asks why.



CAM-07 · Platform 1 · keyframe at the surge (stock footage, faces masked)

DETECTOR + RULE

crowd_surge · people 7 → 11 within 6 s

QWEN2-VL-2B · 336×336 · ON-DEVICE

Q: Is a train stopped at the platform with passengers getting on or off?

"Yes."

OUTCOME

Retitled **"Crowd surge — likely routine train arrival"**, kept at low priority, reason written to the audit log.

Guardrail: the VLM may only **downgrade camera-only** incidents. It can never override an alarm or a human report — the officer can always override it.

Measured on the prototype (laptop, same model and input size as the Hailo-10H HEF). Stage II measures it on the UGen300.

Four systems, nine seconds, one incident card.

17:36:07 **DET-B2** · Platform 2 · emergency intercom pressed (alarm I/O)

17:36:07 **CAM-03, CAM-05** · snapshots pulled on demand (YOLO people count)

17:36:15 **RADIO-1** · Whisper transcript

"Control, this is platform 2. A group of people are running toward exit C, someone may be injured. Requesting backup."

17:36:16 **Fused** → HIGH · "Alarm corroborated by staff report" · zones B, C · 5 signals · checklist attached

RECOMMENDED STEPS (EDITABLE PER SITE)

1 Open linked clips · 2 Call back the reporting staff · 3 If injury confirmed, notify station master & emergency services · 4 Consider holding trains / alternate exits



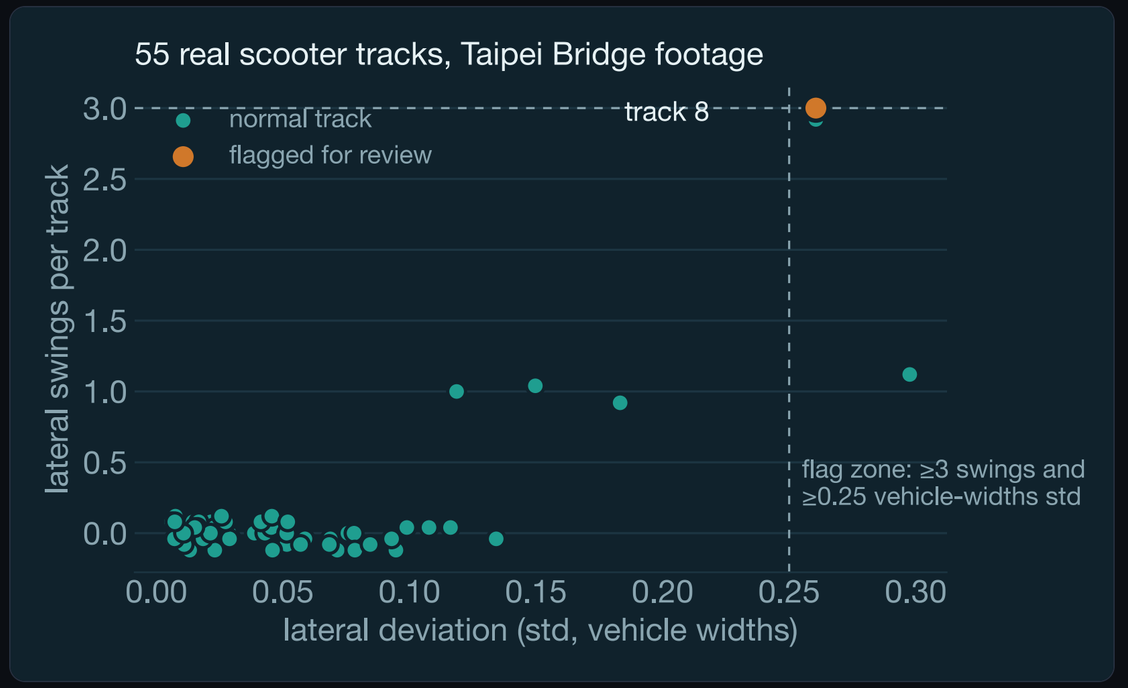
Whisper routes by the last place mentioned — the caller stands on Platform 2 but reports exit C, so the signal lands in zone C, adjacent to the alarm. Radio audio is transcribed on-box and discarded.

Virtual station stitched from openly licensed stock clips (Pexels); detector alarm and radio call are scripted inputs, everything after them is the real pipeline output.

55 real scooter tracks. One worth a second look.



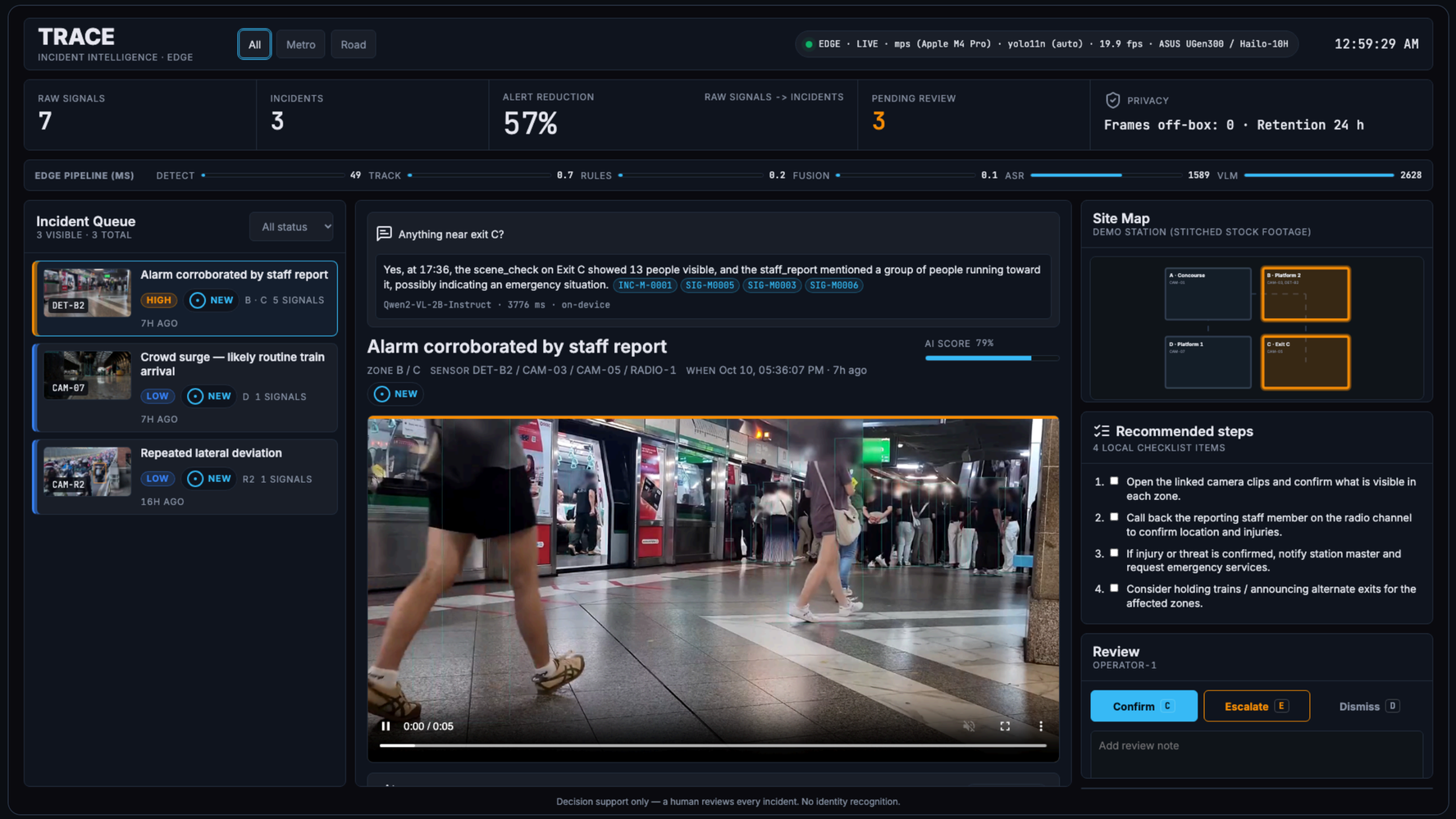
Per track: project motion onto the local flow, remove lane curvature with a quadratic fit, and measure the residual in **vehicle widths** — scale-free across camera heights. Count lateral swings with hysteresis.



- 98% of tracks are smooth; the flag rate sets the human workload.
- Thresholds are per-camera percentiles, calibrated in Stage II with labelled clips.
- The VLM checks for an obstacle ahead → "possible avoidance maneuver".

Footage: June 2023 motorcycle waterfall on Taipei Bridge — Tze Chiang Hao, CC BY-SA 4.0 (Wikimedia Commons). A flag is a review request, never a judgement about the rider.

Everything about an incident on one screen — and you can ask it questions.



Ask the station

"Anything near exit C?" → on-device LLM answers from the event log only, with clickable citations.

Evidence-linked

Every sentence of the summary cites a signal ID.

One-key review

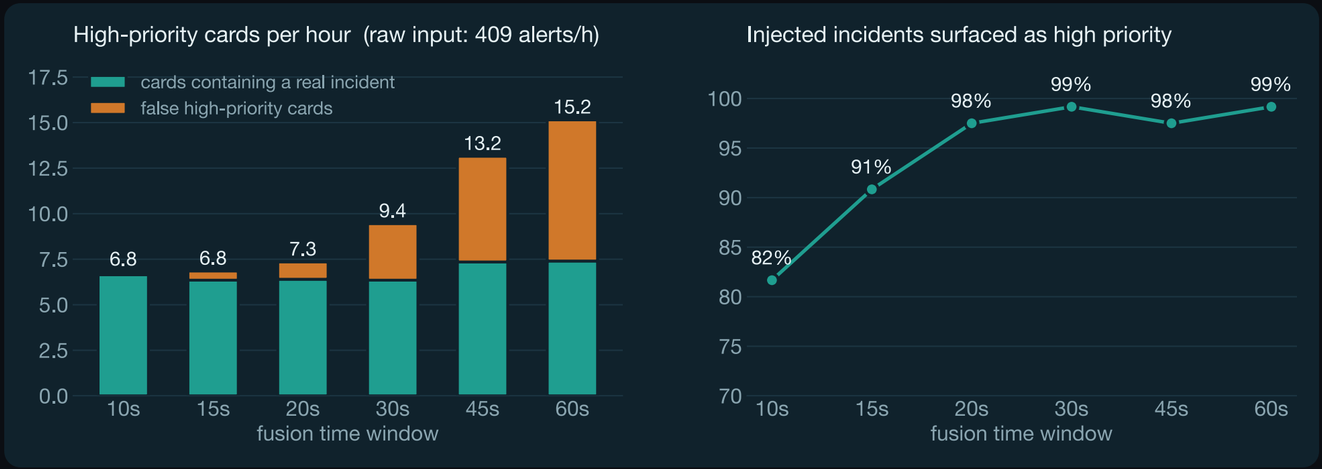
C / E / D to confirm, escalate, dismiss; audited.

Privacy visible

"Frames off-box: 0 · retention 24 h".

Screenshot of the running prototype (FastAPI + vanilla JS) replaying the metro and road runs.

409 alarms an hour become ~9 cards — and 99% of real incidents are still caught.



Synthetic stress test, 20 seeds: 16-zone station, 32 cameras, 16 detectors, 4 radios; 1 h of nuisance alarms (6/min) + 6 injected multi-zone incidents.

PER HOUR	CARDS	RECALL
Every alarm is a card	409	100%
Per-sensor de-dup (VMS-style)	373	100%
TRACE high priority	9.4	99%

OFFICER TIME, PER HOUR

102 min
glance at every alarm (15 s each)

14 min
TRACE: 60 s per high + 15 s per medium card

At a 30 s window: 3.1 false high cards/h. A 20 s window trades 1% recall for 1 false card/h — tunable per site.

Reproduce: python scripts/simulate_alert_storm.py --seeds 20 --window 30 · triage-time figures are stated assumptions, to be measured with operators in Stage II

What runs today, how fast, and what changes on the UGen300.

STAGE	PROTOTYPE (MEASURED)	UGEN300 (PLANNING)
YOLO11n detect, 960 px	49 ms / frame	~3.3 ms (302 FPS)
ByteTrack + rules (host)	0.9 ms / frame	same host code
Fusion (host)	0.08 ms / run	same host code
Whisper, 8.7 s radio call	1,589 ms	Whisper-Base 25 tok/s
Qwen2-VL-2B, verify + describe	3,122 ms / incident	TTFT 1.0 s · 7 tok/s

Prototype: mps (Apple M4 Pro), PyTorch MPS, unoptimised. UGen300 column: Hailo Model Zoo figures (PCIe); USB numbers are the first Stage II measurement.

BOX BUDGET · 8 CAMERAS @ 15 FPS

120 detections/s → **40%** of YOLOv11n capacity on the Hailo-10H; Whisper and the VLM fill gaps only on events (~1 incident/min at peak → VLM busy < 20% of the time).



ENGINEERING EVIDENCE

- 15 unit tests (fusion, VLM verifier, trajectories, ask)
- Codex-assisted code review: 11 issues found & fixed
- Reproducible runs: one YAML per scenario

Stage timings from data/run_metro.json · data/run_road.json (pipeline.stage_ms)

Safety depends on knowing what the system must not do.

No identity, ever

No face recognition, no cross-camera re-identification. Head regions are blurred in every stored frame before it is written.

Never infer intent or intoxication

Prompts forbid guessing identity, emotion or intent; road flags describe motion only and always go to human review.

Nothing raw leaves the box

Video and radio audio are processed in memory; only masked 6-s clips attached to incidents are kept (24 h default) and only metadata is sent upstream.

Known limits, stated up front

Short or occluded tracks can't support a conclusion; a crowd surge without context stays low; thresholds are site-calibrated, not universal.

The model can't escalate itself

The VLM may only downgrade camera-only incidents; every model verdict and human decision is in the audit log.

Aligned with Taiwan PDPA

Purpose-bound processing, data minimisation and retention limits are defaults, not options.

Streaming every camera to the cloud costs more than the hardware.

PER CAMERA, PER MONTH	CLOUD ANALYTICS	TRACE EDGE
Uplink (1080p H.264 ≈ 4 Mbps)	≈ 1,296 GB	< 0.1 GB metadata
Ingest (AWS Kinesis Video, $0.0085/GB)	≈ US$11	—
Analysis (Rekognition streaming events, $0.00817/min)	≈ US$353	—
Accelerator energy (2.5 W ÷ 8 cameras)	—	≈ 0.23 kWh
Raw video off-site	yes	no

>99.9%

less upstream bandwidth: events, not video.

2.5 W

fanless accelerator; fits in an existing equipment cabinet or NVR.

Also works when the network is down — exactly when incidents happen.

Sources: AWS Kinesis Video Streams & Rekognition pricing pages (accessed 2026-10-10); bitrate rule of thumb (Nelly's Security, 2026). Cloud figures for continuous analysis of one stream.

Land with one station. Expand through the cameras already installed.

GLOBAL VIDEO ANALYTICS

US$22.6B

by 2028, 22.3% CAGR (MarketsandMarkets)

TAIWAN BEACHHEAD

Taipei Metro alone is adding **2,901 AI cameras** across 31 stations; Taichung runs ~32,000 public cameras (≈80% outdated), Tainan 19,443.

MODEL	HOW
Edge box	Host + UGen300 sold through system integrators / ASUS IoT channel
Software licence	Annual, per camera; includes rule packs (metro, road, campus, factory)
Pilot	Paid 8-week pilot: one station, KPI = officer minutes per verified incident
Expansion	Other rail & bus operators, city traffic units, campuses, logistics sites

Buyer: the operator who owns the incident-response workflow. TRACE complements their VMS and AI cameras via webhook/JSON — no rip-and-replace.

Sources: MarketsandMarkets IVA report (press release); Taiwan News 2026-08-21; UDN 2026-04-29; CTEE 2026-06-01. Pricing to be validated with pilot customers.

Others detect, search or correlate in the data centre. TRACE fuses and verifies at the edge.

PRODUCT	WHAT IT DOES	WHERE IT RUNS	SPEECH	VLM VERIFY
Genetec Mission Control	Correlates alarms into incidents with procedures	Server / VMS	—	—
ZeroEyes (e.g. SEPTA)	Gun detection, human-verified alerts	Cloud ops centre	—	human
Acusensus Heads-Up	Flags possibly impaired drivers for officers	Roadside system	—	—
Milestone Hafnia (2025 hackathon)	VLM adds context to edge events	Cloud VLM	—	✓
TRACE	**Fuses camera + alarm + radio, verifies with a VLM, evidence-linked cards**	**2.5 W on-site**	✓	✓

The gap is the combination: multi-source fusion + speech + VLM verification, running offline on one low-power accelerator next to the cameras.

Sources: Genetec resources; Route Fifty 2022-12-02; Road Safety GB 2025-01-29; Milestone press release (Hafnia Hackathon 2025)

Four weeks to move every number onto the UGen300.

WEEK 1 · BRING-UP

HEFs on the stick

YOLOv8s/11n, Whisper-Base, Qwen2-VL-2B on one shared VDevice; measure USB FPS, latency, watts.

KPI

≥ 8 cameras @ 15 fps on one UGen300

WEEK 2 · METRO

Live multi-source

2–4 RTSP cameras + GPIO "alarm" + live radio mic; end-to-end alarm-to-card latency.

KPI

Alarm → card < 2 s; VLM verdict < 5 s

WEEK 3 · ROAD + VLM

Calibrate & verify

Label ~200 public road clips; per-camera thresholds; VLM second-opinion precision.

KPI

≥ 90% incident recall at ≤ 2 false high cards/h

WEEK 4 · VALIDATE

Operator study

5 participants triage a replayed alarm hour with vs without TRACE; README, model card, benchmarks.

KPI

Triage time −50% in the operator study

Main risks → mitigation: USB throughput lower than PCIe → YOLOv11n + lower fps on static cameras · VLM latency → only per incident, async · few real incidents → replayed public datasets + staged tabletop scenes · Mandarin radio → Whisper-Small + keyword lexicon.

TRACE starts with evidence.

TEAM (SOLO)

Wei-Lun (Alen) Hsiao

B13902049 · CSIE, National Taiwan University

TRACK · THEME

Battlefield Lightning · ASUS UGen300

Workplace AI (station & traffic operations)

GITHUB REPOSITORY

github.com/AlenHsiaoWeiLun/trace-ugen

DEMO VIDEO (YOUTUBE)

youtu.be/ — link added after upload

Full references and media credits in the appendix. All footage is openly licensed stock/public-domain material; no real incident footage is used. Development assisted by AI coding tools (Claude Code, OpenAI Codex).

APPENDIX A · REFERENCES

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[2] Taiwan News, "Taipei Metro AI camera rollout", 2026-08-21. taiwannews.com.tw/en/news/6425898

[3] Taipei Metro — ridership 2025 (767M trips, 2.10M/day, 118 stations). en.wikipedia.org/wiki/Taipei_Metro

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[10] Hailo Model Zoo v5.4, HAILO10H object detection benchmarks. github.com/hailo-ai/hailo_model_zoo

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[12] hailo-ai/hailo-apps — simple_vlm_chat, simple_whisper_chat (API reference). github.com/hailo-ai/hailo-apps

[13] AWS Kinesis Video Streams pricing; Amazon Rekognition pricing (accessed 2026-10-10).

[14] Nelly's Security, camera bitrate guide, 2026-04-17.

[15] MarketsandMarkets, Video Analytics Market (US$22.6B by 2028, 22.3% CAGR).

[16] UDN 2026-04-29 (Taichung cameras); CTEE 2026-06-01 (Tainan cameras).

[17] Road Safety GB, Acusensus trial in Devon & Cornwall, 2025-01-29; Vision Zero South West.

[18] Route Fifty, "Transit agency pilots gun detection tech" (ZeroEyes, SEPTA), 2022-12-02.

[19] Genetec Mission Control, key features and benefits.

[20] Milestone Systems, Hafnia Hackathon 2025 winners press release.

[21] Wang, Y. et al., Qwen2-VL, 2024; Radford, A. et al., Whisper, 2022; Zhang, Y. et al., ByteTrack, ECCV 2022; Ultralytics YOLO11, 2024.

[22] Taiwan Personal Data Protection Act (PDPA).

APPENDIX B · MEDIA CREDITS & DATA PROVENANCE

ASSET	AUTHOR / SOURCE	LICENSE	USED FOR
Platform 1 & 2, MRT Taipei Main Station (2026)	Bigmorr, Wikimedia Commons	CC BY-SA 4.0	pp. 1, 20
People at Bannan Line Platform (2010)	Bernard McManus, Wikimedia Commons	CC BY 2.0	p. 2
Motorcycle waterfall on Taipei Bridge 05/06/07 (2023)	Tze Chiang Hao, Wikimedia Commons	CC BY-SA 4.0	p. 4, road scenario video
Station clips 37737804, 34647618, 35767994, 25786069	Pexels contributors	Pexels License	metro scenario video
Motorcycles traffic in the city (Taiwan) 4740223	Allen Low, Pexels	Pexels License	road scenario video
UGen300 and Hailo-10H product renders	ASUS UGen Series Sales Kit	contest material	p. 6
Radio call audio	Synthesised (neural TTS) + band-pass "radio" filter	generated	metro scenario

What is real and what is staged. Detection, tracking, rule signals, Whisper transcription, fusion, VLM verdicts, evidence clips and all timings are real outputs of the prototype in the GitHub repository. The metro "station" is stitched from stock clips of different stations; the detector alarm and the radio call are scripted inputs. The alert-storm benchmark uses synthetic alarms. No footage of real incidents is used.